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COMMON QUARTERLY EXAMINATION - 2026

Standard - 12 Reg.No. [| | | | | |]

Time Allowed: 3.00 Hours **BIOLOGY** *Maximum Marks: 70*

PART-I
BIO- BOTANY (Marks: 35)
SECTION-1

Note: Choose the correct answer: **8×1=8**

1) The genes controlling the seven pea characters studied by Mendel are known to be located on how many different chromosomes?
   a) Seven b) Six c) Five d) Four
2) Which of the following represents megagametophyte?
   a) Ovule b) Embryosac c) Nucellus d) Endosperm
3) Plasmids are
   a) circular protein molecules b) tiny bacteria
   c) can't replicate by themselves
   d) contain marker for antibiotic resistance
4) Due to incomplete linkage in maize, the ratio of parental and recombinants are
   a) 50:50 b) 7:1:1:7 c) 96:4:3:6 d) 1:7:7:1
5) Solidifying agent used in plant tissue culture is
   a) Nicotinic acid b) Cobaltous chloride
   c) Calcium chloride d) Agar
6) The two genes that are sufficiently far apart on the same chromosome are called
   a) Bivalent b) Syntenic genes c) Multiple alleles d) Linked genes
7) The genotypic ratio of Dihybrid test cross is
   a) 1:1:1:1 b) 9:3:3:1 c) 1:2:1 d) 12:3:1
8) An analysis of chromosomal DNA using the Southern Hybridisation technique does not use
   a) Electrophoresis b) Blotting
   c) Autoradiography d) Polymerase Chain Reaction

SECTION-2

Note: Answer any Four of the following: **4×2=8**

9) Distinguish tenuinucellate and Crassinucellate ovules.
10) Give the names of the scientists who rediscovered Mendelism.
11) What are called linked genes?
12) What is gene mapping?
13) Define - Bioremediation.
14) Enumerate the disadvantages of Bt Cotton.

SECTION-3

Note: Answer any three of the following. Q.No:19 is Compulsory: **3×3=9**

15) What is known as Atavism?
16) Draw the structure of an Ovule and label the parts.
17) Mention any three importance of crossing over.
18) What do you know about the word pBR^{322}?
19) What is Cryopreservation?

SECTION-4

Note: Answer all questions:

20) Describe Incomplete dominance with an example. **2×5=10**
    Mention the applications of Biotechnology. **(OR)**
21) Explain the different types of Ovules.
    Explain the mechanism of Crossing Over. **(OR)**

PART-II
BIO-ZOOLOGY (Marks:35)
SECTION-1

Note: Choose the correct answer:

1) XO type of sex determination and XY type of sex determination are examples of **8×1=8**
   a) Male heterogamety b) Female heterogamety
   c) Male homogamety d) Both (b) and (c)

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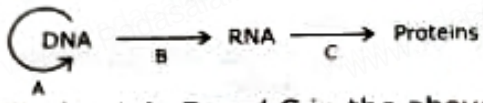
2) The Skene's glands of female are homologous to
a) Bulbourethral glands of the male b) Prostate gland of male
c) Cowper's gland of male d) Glans penis of male

3) Match the following:
1) Trichoderma polysporum - a) Statins
2) E. coli - b) Citric acid
3) Monascus purpureus - c) Cyclosporin A
4) Aspergillus niger - d) Recombinant human insulin
a) 1-d, 2-c, 3-b, 4-a b) 1-b, 2-d, 3-a, 4-c
c) 1-d, 2-a, 3-c, 4-b d) 1-c, 2-d, 3-a, 4-b

4) Causing a mud ulcer in humans
a) Virus b) Protozoa c) Fungus d) Bacteria

5) The mode of sexual reproduction in bacteria is by
a) Formation of gametes b) Endospore formation
c) Conjugation d) Zoospore formation

6) Which of the following is the contribution of Hugo De Vries?
a) Theory of Inheritance of acquired characters
b) Mutation theory
c) Modern Synthetic theory d) Theory of Natural Selection

7) 

Find out A, B and C in the above pictorial representation.
a) A - Transcription, B - Translation, C - Replication
b) A - Translation, B - Transcription, C - Replication
c) A - Replication, B - Translation, C - Transcription
d) A - Replication, B - Transcriptions, C - Translation

8) Identify the correct statements from the following:
a) Chlamydiasis is viral disease
b) Both syphilis and gonorrhoea are easily cured with antibiotics
c) Gonorrhoea is caused by a spirochaete bacterium, Treponema palladium
d) The incubation period for syphilis is 2 to 14 days in males and 7 to 21 days in females

Section-2

Note: Answer any Four of the following: **4×2=8**

9) List the major gases seems to be found in the primitive earth.
10) What are interferons?
11) Differentiate between External and Internal fertilization.
12) Why human genome project is called as mega project?
13) How is polyspermy avoided in humans?
14) Define - Zymology.

Section-3

Note: Answer any three questions. Q.No.19 is compulsory: **3×3=9**

15) Write short notes on Phases of life cycle.
16) Differentiate between divergent evolution and convergent evolution with one example each.
17) Give the applications of DNA finger printing.
18) What is Bioremediation? Give one example for the same.
19) What are the steps used to reduce the risk bacterial resistance?

Section-4

Note: Answer the following questions: **2×5=10**

20) List the salient features of Genetic code. **(OR)**
If immunological surveillance is effective, causes should not occur. Justify the statement.

21) Explain the various phases of menstrual cycle. **(OR)**
What is referred to as Industrial Alcohol? Discuss its preparation.

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